Filtered Smith Predictor for Sub-Pixel Tip-Tilt Stabilization

Mathematical basis, causal limits, code architecture, measured attenuation results, failure analysis, latency budget, and next-step hardware strategy.

Snapshot	Value
System	ASI290MM, 64x64 ROI, tracked 15x15 centroid, 2309-2310 Hz
Plant	Ky=0.300, Kx=0.427 normalized error per command unit
Effective delay	1.81 ms = 4.18 frames (current FSP model)
Best completed FSP run	attenuation11: 0.127 px Y, 0.218 px X closed RMS
Linear bound, run11 PSD	0.0769 px Y, 0.1203 px X at 4.18 frames
Run14	Frozen order-128 predictor with corrected safety logic; running at report time

Executive finding

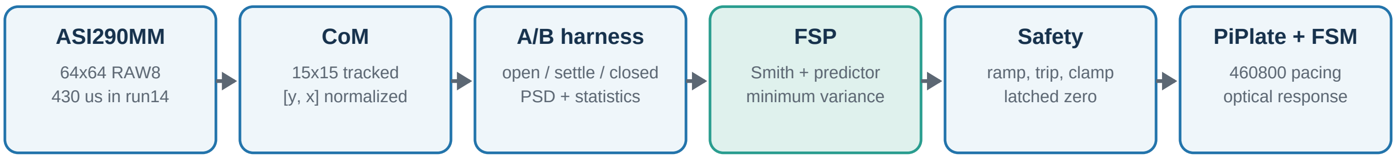
The disturbance is highly predictable in variance (about 96-97% under the scalar linear model), but the remaining innovation is still 17-19% in RMS. The current controller is limited by model mismatch, closed-loop identification safety, and the 1.81 ms causal horizon. A verified latency reduction is likely to buy more than a more complex model using the same centroid stream.

Status and scope

This report is a workspace snapshot through 2026-07-16. Attenuation10 and attenuation11 are completed measured runs. Attenuation12 produced two hardware instabilities. Attenuation13 false-tripped at closure. Attenuation14 is configured and running, so no run14 performance number is claimed here. Derived Wiener bounds and hardware projections are explicitly labeled.

Contents: system model; estimator/control mathematics; PSD and conditional-MMSE bounds; software architecture; run findings; failure analysis; latency, exposure, ROI, and soundcard options; telemetry needed for the true conditional MMSE; references and artifact map.

1. System, coordinates, and measured data



Current attenuation pipeline

Figure 1. Current attenuation pipeline and the internal command path used by anyloop:fsp.

Coordinate convention

The camera centroid is emitted in normalized image coordinates. For an N-pixel axis, pixel position p and normalized coordinate c satisfy:

```
c = -1 + 2 p/(N-1) p = (N-1)(c+1)/2
```

For the 64x64 ROI, one normalized output unit equals (64-1)/2 = 31.5 px. Jitter conversions therefore use sigma_px = 31.5 sigma_norm. The run statistics are RMS about the mean, so static pointing offsets are excluded from the jitter score.

PSD, attenuation, and band RMS

```
sigma_band = pixel_scale * sqrt( integral_[f1,f2] S(f) df )
A_dB(f) = 10 log10( S_closed(f) / S_open(f) )
```

Negative A_dB means attenuation; positive A_dB means amplification. Welch PSDs use a Hann window and 50% overlap in attenuation_test. Open and closed phases use their own measured sample rates and are compared on a shared frequency grid.

Current configuration snapshot

Parameter	Attenuation14	Steering config
Camera exposure	430 us	200 us
Centroid threshold	30	22
FSP order / NLMS mu	128 / 0.005	128 / 0.005
Delay	4 + 0.18 frames	4 + 0.18 frames
Identification	500 s open, then frozen	20 s startup, then frozen
Authority ramp	30 s	15 s
Safety	error envelope + 0.05; |u|>0.65; 8 frames	same

Important configuration split

The long attenuation14 test still uses 430 us exposure and threshold 30. The separate steering_fsp.json currently uses 200 us and threshold 22. Results from those configurations should not be mixed without a new Bode gain, latency, and noise measurement.

2. Plant model and Smith reconstruction

Delayed plant

```
e(k) = phi(k) + K u_p(k-d-f)
```

Here e is measured residual, phi is the disturbance that would be seen open loop, K is the signed command-to-error gain, d is the integer delay, and f is the fractional remainder. Current values are d=4, f=0.18, Ky=0.300, and Kx=0.427. The sign of K is safety-critical because the minimum-variance command uses -1/K.

Fractional delay: first-order Thiran all-pass

```
a = (1-f)/(1+f) u_f(k) = a u(k) + u(k-1) - a u_f(k-1)
```

The Thiran section gives unity magnitude and the requested low-frequency group delay. It avoids the artificial high-frequency magnitude droop of linear interpolation, which otherwise appears as leaked command in the Smith reconstruction.

Smith reconstruction

```
phi_meas(k) = e(k) - K u_f(k-d)
```

Under an exact plant model, phi_meas is independent of the controller's previous command. During the startup hold u=0, so phi_meas is exactly the raw open-loop disturbance and is the cleanest identification signal.

Observed failure mechanism	If K or delay is wrong, command leakage remains in phi_meas. Continuous closed-loop NLMS can then learn its own command as disturbance, creating a positive-feedback path. This matches attenuation12: one instability recovered and a later event spiraled out of control.

Startup authority

```
alpha(t) = 0, t=t_close
u_requested(k) = -alpha(k) phi_hat(k+d+f | I_k) / K
```

The ramp produces a bumpless handoff. Output is then optionally filtered, hard-clamped to [-1,1], and passed to the actuator bridge.

3. Modal AR(2) and stationary Kalman estimator

Per-vibration AR(2) model

```
x_i(k) = a1_i x_i(k-1) + a2_i x_i(k-2) + nu_i(k), Var(nu_i)=q_i
a1_i = 2 exp(-2 pi zeta_i f_i Ts) cos(2 pi f_i Ts sqrt(1-zeta_i^2))
a2_i = -exp(-4 pi zeta_i f_i Ts)
```

The stacked state X=[x_i(k), x_i(k-1)] contains up to eight modes per axis. The measurement matrix C sums the current position component of every mode. Configured sharp-line damping ratios are approximately 0.002-0.003; a zeta=0.30 mode represents the broad 20 Hz continuum.

Stationary variance scaling

```
Gv_i = (1-a2_i) / [(1+a2_i)(1-a1_i-a2_i)(1+a1_i-a2_i)]
```

Gv converts drive variance q into visible modal state variance. For lightly damped modes it is approximately 1/(4 pi zeta f Ts), often thousands to tens of thousands. Feeding measured state energy directly into q inflates q/r by orders of magnitude and reproduces the high-frequency waterbed.

Kalman covariance and gain

```
P_minus = A P_plus A^T + Q
L = P_minus C^T / (C P_minus C^T + r)
P_plus = (I - L C) P_minus
X_pred = A X_hat X_hat = X_pred + L[phi_meas - C X_pred]
```

fsp_solve_gain iterates this Riccati recursion at initialization and on each modal adaptation tick. The hot path is only O(number of modes): predict, form the scalar innovation, and correct with the precomputed stationary gain.

Delay-step modal prediction

```
phi_hat(k+d|k) = C A^d X_hat(k|k)
```

Optional command low-pass compensation rolls each modal state forward additional integer steps equal to the filter phase lag at its center frequency and applies a gain correction capped at 3. The current configuration leaves this command filter disabled because simulation showed that it costs prediction horizon without removing the underlying waterbed mechanism.

4. Full-band Wiener/FIR observer and safe identification

Causal FIR predictor

```
x_k = [phi_meas(k), phi_meas(k-1), ..., phi_meas(k-P+1)]^T
phi_hat_d(k) = w_d^T x_k
```

With broad_order>0, this full-band prediction supplies the command. The AR(2) modal estimate remains available for diagnostics/adaptation but is not summed, because both paths estimate the same disturbance and summing would double-cancel it.

Normalized LMS identification

```
epsilon_k = phi_meas(k) - w_k^T x_(k-d)
w_(k+1) = w_k + mu epsilon_k x_(k-d) / [delta + ||x_(k-d)||^2]
```

Attenuation11 used P=512 and mu=0.03. The robust configuration uses P=128 and mu=0.005 to reduce coefficient misadjustment and high-frequency overfit. Two adjacent predictors estimate horizons d and d+1; the fractional prediction is blended as:

```
phi_hat_(d+f) = (1-f) phi_hat_d + f phi_hat_(d+1)
```

Freeze policy

The safe policy trains only while the startup hold guarantees u=0, then freezes the broadband weights before authority rises. Modal diagnostic statistics may continue to update, but the active broadband command law does not adapt in closed loop.

Figure 2. Attenuation14 identification and validation lifecycle.

Safety detector

```
trip_error if |e(k)| > |e_open_baseline| + 0.05 for 8 consecutive frames
trip_command if |u_requested(k)| > 0.65 for 8 consecutive frames
```

The latch forces command to zero until restart. The open baseline is an EWMA learned during the hold. This corrects attenuation13's false trip: its legitimate X operating point was about -0.133 normalized units (-4.19 px), already above the original absolute 0.05 error threshold. Static X recentering needs about 0.133/0.427 or 0.31 command units, so the original 0.35 command trip left only about 0.04 units for ordinary dynamic correction.

5. Causal MMSE, Wiener bound, and predictable variance

True conditional-MMSE theorem

```
g*(I_k) = E[x(k+d) | I_k]
MMSE(d) = inf_g E[(x(k+d)-g(I_k))^2] = E[ Var(x(k+d) | I_k) ]
```

No causal model can beat this value using the same information set I_k and the same delay. A more powerful model can beat the scalar linear result only by exploiting information excluded from it: command history, X/Y coupling, raw spot shape, nonlinearities, or nonstationarity.

What is and is not known	The true numerical conditional MMSE was not measured. From the PSD alone, the rigorous relation is 0 <= true MMSE <= linear Wiener MSE. If the relevant signals are jointly Gaussian, the conditional mean is linear and the Wiener result equals the true conditional MMSE.

PSD-to-autocovariance

```
R(l) = integral_[0,fs/2] S(f) cos(2 pi f l/fs) df
```

Wiener-Hopf equations for a P-tap predictor

```
R_P w = r_d, [R_P]_(i,j)=R(|i-j|), [r_d]_i=R(d+i)
MSE_linear(d) = R(0) - r_d^T R_P^(-1) r_d
sigma_px(d) = 31.5 sqrt(MSE_linear(d))
```

The workspace utility uses the one-sided measured open-loop PSD, direct fractional-lag cosine integration, P up to 512, and includes centroid measurement noise. It is an infinite-data stationary linear bound estimated from a finite Welch PSD, not an unconditional physics limit.

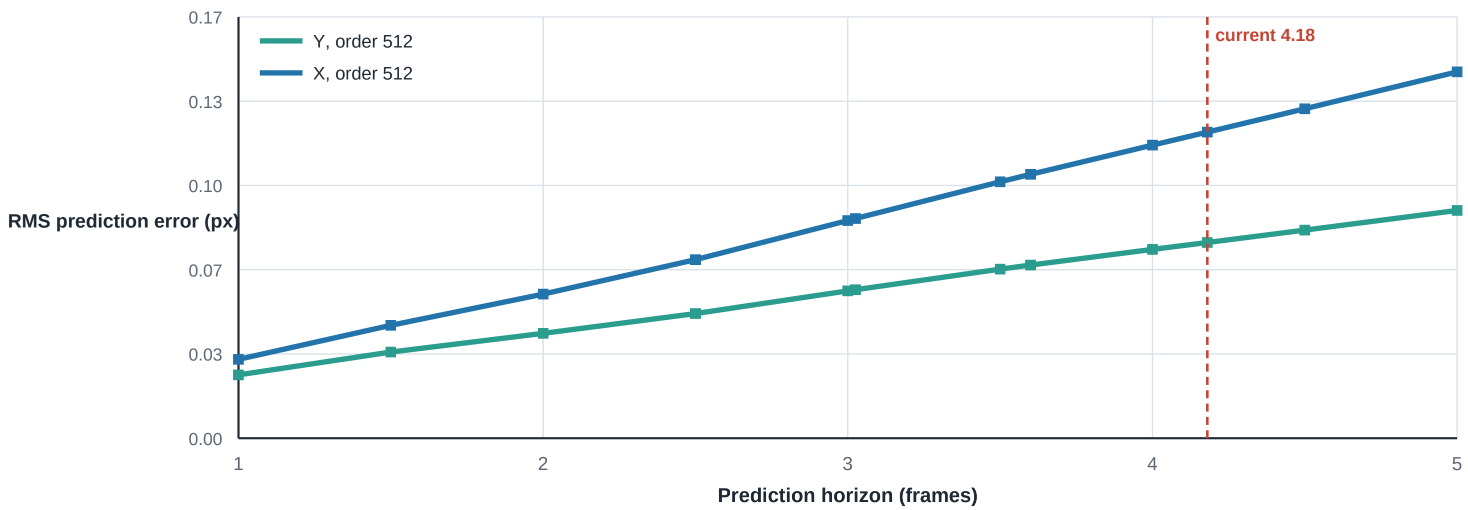


Figure 3. Attenuation11 scalar linear prediction floor versus causal horizon.

6. Measured attenuation findings

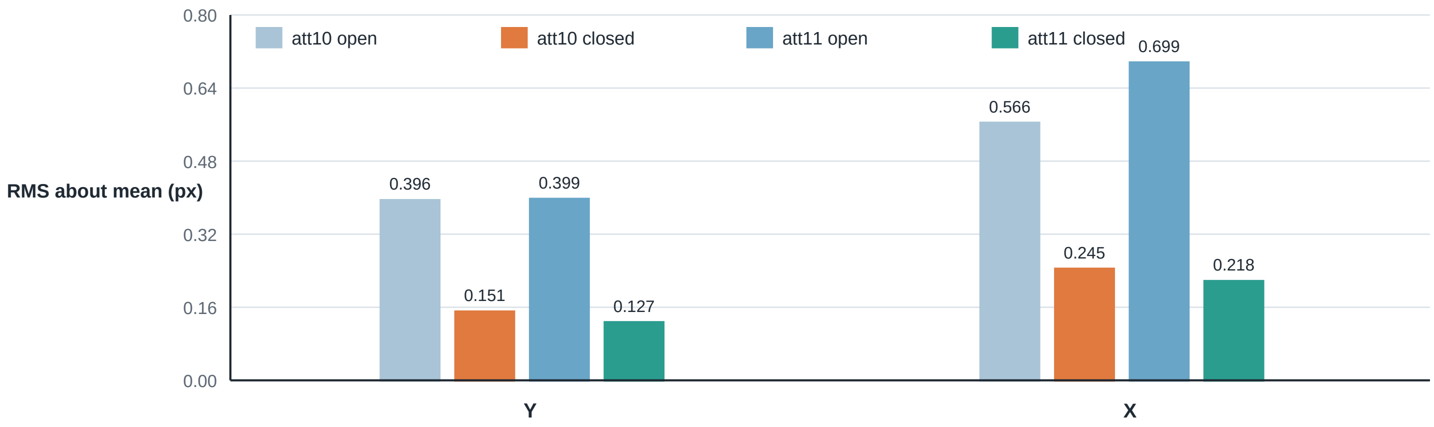


Figure 4. Measured RMS-about-mean. Different open-loop environments make closed-loop A/B interpretation more reliable than raw open RMS comparison.

Metric	Attenuation10 PID+lead	Attenuation11 FSP	Change in closed RMS
Y open / closed	0.396 / 0.151 px	0.399 / 0.127 px	15.5% lower
X open / closed	0.566 / 0.245 px	0.699 / 0.218 px	11.0% lower
Y mean open / closed	+3.778 / +0.000 px	-0.044 / +0.001 px	pointing, not jitter
X mean open / closed	-7.494 / +0.000 px	-4.191 / +0.000 px	run13 trip clue

Attenuation11 improved closed RMS by 15.5% in Y and 11.0% in X despite a substantially worse X open-loop disturbance. It strongly rejected the 0-64 Hz structure but paid for that suppression with a high-frequency waterbed.

Attenuation11 band RMS (px)

Band (Hz)	Y open	Y closed	X open	X closed	Interpretation
0-10	0.1097	0.0256	0.2574	0.0602	strong rejection
10-20	0.1524	0.0312	0.2836	0.0582	strong rejection
20-30	0.2674	0.0317	0.3937	0.0561	strong rejection
30-40	0.0482	0.0252	0.0632	0.0551	strong rejection
40-50	0.0338	0.0258	0.0704	0.0547	strong rejection
50-64	0.1229	0.0304	0.1347	0.0637	strong rejection
64-90	0.0352	0.0397	0.1282	0.0857	transition
90-120	0.0538	0.0411	0.0585	0.0904	transition
120-200	0.0432	0.0659	0.0600	0.1025	waterbed
200-400	0.0297	0.0564	0.0214	0.0206	waterbed
400-1155	0.0123	0.0265	0.0120	0.0254	waterbed

7. Spectral evidence from attenuation11

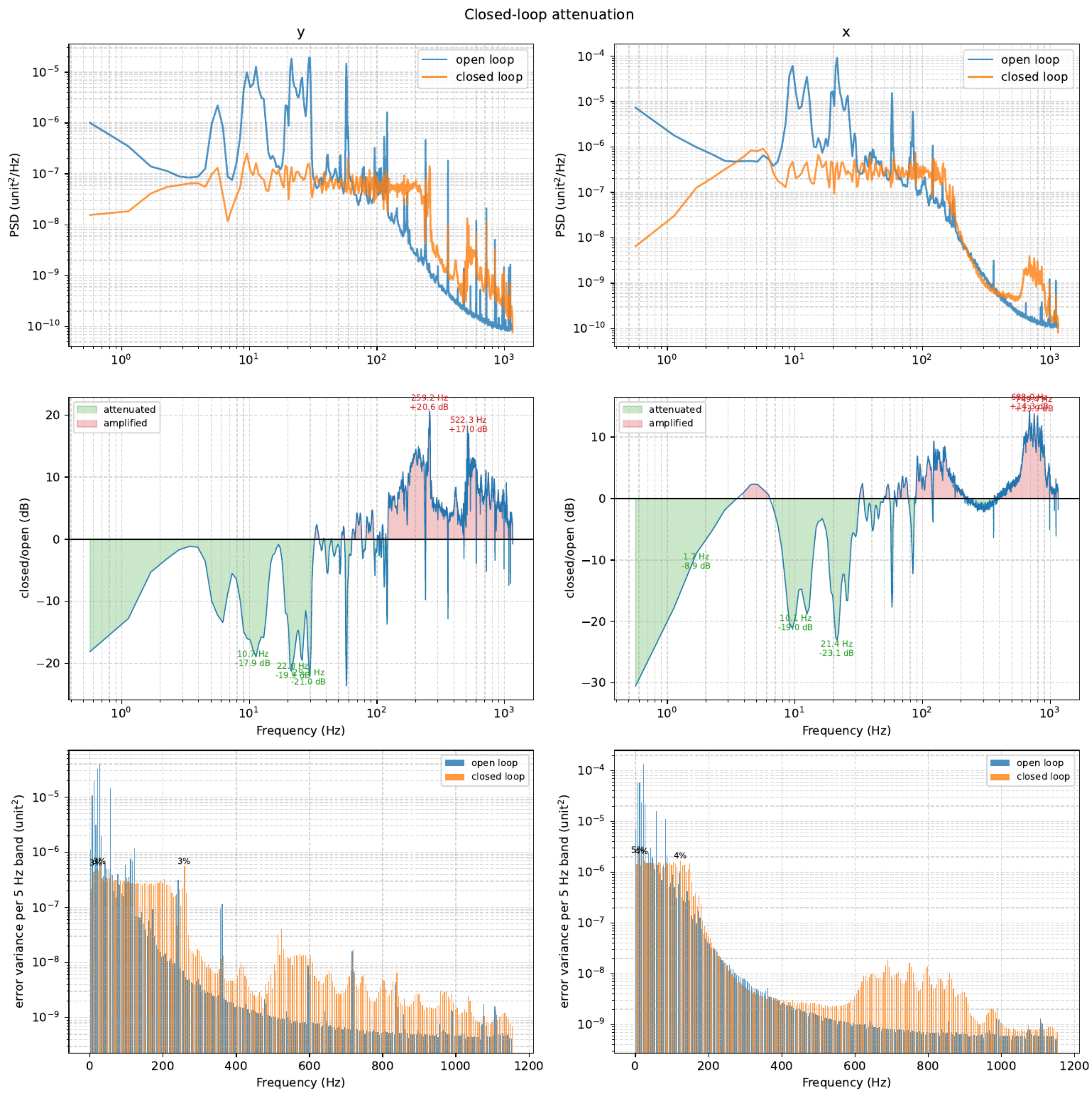


Figure 5. Original attenuation11 report page. Top: open/closed PSD. Middle: attenuation in dB. Bottom: 5 Hz-band variance. The 120-400 Hz and higher amplification is visible directly.

Spectral finding	Low-frequency rejection is excellent: prominent 10-30 Hz lines fall by roughly 18-23 dB. Above the useful band the controller amplifies residuals, with observed local peaks near +20 dB in Y around 259-260 Hz and about +14 dB in X around 688-794 Hz. This is why total RMS cannot be judged from line suppression alone.

8. Prediction limits and interpretation

Quantity	Y	X	Meaning
Open RMS, attenuation11	0.3986 px	0.6986 px	measured
Order-512 bound, d=4.18	0.0769 px	0.1203 px	derived scalar linear floor
Order-128 bound, d=4.18	0.0836 px	0.1261 px	robust model class
Predictable variance	96.3%	97.0%	1 - residual variance/open variance
Residual RMS fraction	19.3%	17.2%	square root makes small variance visible
Actual att11 / bound	1.66x	1.81x	implementation/model gap

Predictable fraction formula

```
F_predictable = 1 - sigma_prediction^2 / sigma_open^2
```

Most variance is predictable because strong low-frequency lines dominate the open spectrum. Yet 3-4% remaining variance corresponds to 17-19% of open-loop RMS. That remaining innovation includes high-frequency continuum, sensor noise, line drift, and changes over the 1.81 ms horizon.

Attenuation10 versus attenuation11 PSD bounds

Open PSD source	Delay	Y bound	X bound
attenuation10	4.00	0.0617	0.1026
attenuation10	4.18	0.0644	0.1071
attenuation10	5.00	0.0768	0.1278
attenuation11	4.18	0.0769	0.1203

The often-quoted 0.0644 px Y / 0.1071 px X values came from the attenuation10 open PSD. The attenuation11 environment was harder, especially in X, so its own bound is 0.0769 / 0.1203 px. Neither pair is a universal hardware limit.

Waterbed relation

```
R(omega) = 1 - exp(-j omega d) T(omega)
integral log |R(omega)| d omega = 0 (stable causal conditions)
```

Deep rejection in one band must be paid for elsewhere for a stable causal sensitivity-like transfer. Prediction changes where the trade is paid by exploiting disturbance structure; it does not abolish causality or the innovation floor.

9. Run history and failure analysis

Run	Configuration	Observed result	Lesson / action
10	PID + lead + line oscillators + Kalman	0.151 px Y, 0.245 px X	classical baseline
11	FSP, order512, mu=0.03, closed adaptation	0.127 px Y, 0.218 px X; strong HF waterbed	broad prediction works, but high-order/noisy adaptation is costly
12	order128, mu=0.005, Thiran; still adaptive closed loop	two massive instabilities; second required stop	closed-loop NLMS learned leaked command; freeze active weights
13	frozen training + absolute trip thresholds	tripped immediately at closure	normal X offset exceeded |e|>0.05; command threshold had little dynamic margin
14	open-only training; frozen weights; baseline-relative trip; |u|>0.65	running at report time	await completed PDF/DAT before performance judgment

Safety invariants now implemented

Invariant	Implementation
Known-zero identification	pass_open=true and FSP start_delay=600 s; command is zero during training
Frozen active model	broad_freeze_closed=true; FIR weights stop updating before authority ramp
Bumpless closure	30 s authority ramp in attenuation14
Static-offset tolerant error trip	compare |e| to learned |open baseline| + 0.05
Command headroom	trip at 0.65 for 8 frames; hard clamp remains 1.0
Latched safe state	zero command and reset command-filter states until restart

What run14 can establish

A successful run14 primarily validates stability of frozen identification and corrected safety logic. Because it uses order128 rather than the order512 analysis bound, and the disturbance is nonstationary, it should be compared with both the measured run10 baseline and its own run14 open PSD bound.

10. Latency budget and hardware leverage

Current effective delay

```
Ts = 1/2309 s = 0.433 ms tau = 4.18 Ts = 1.81 ms
```

Component	Estimate	Confidence / interpretation
Exposure midpoint	about 0.215 ms at 430 us	half-integration timing contribution
PiPlate update ZOH	about 0.28 ms mean; up to about 0.56 ms	460800 pacing near 1770 updates/s
Camera readout + USB + retrieval	inside remainder	not separately measured in current geometry
CoM + FSP + scheduling	inside remainder	expected small, but not isolated
DAC + FSM + optical response	inside remainder	included in Bode phase fit
Complete command-to-centroid	1.81 ms	measured/fitted quantity used by FSP

The component estimates overlap; they must not be added again to the 1.81 ms Bode fit. The older five-frame Kalman horizon was a conservative accounting using approximately 3.93 base frames + 0.5 exposure frame + 0.65 DAC-ZOH frame.

Latency scenarios

Scenario	Effective horizon	Y linear floor	X linear floor	Status
Current	4.18 frames / 1.81 ms	0.0769	0.1203	derived from run11 PSD
Full path -0.25 ms	about 3.60 frames	0.0680	0.1037	32x32 historical latency gain proxy
Full path -0.50 ms	about 3.025 frames	0.0583	0.0863	upper hardware scenario
Fast branch only >20 Hz	mixed	about 0.063	about 0.098	engineering interpolation, not a bound

Phase cost of latency

```
Delta phase(deg) = 360 f Delta_tau
```

A 0.5 ms reduction removes 3.6 deg at 20 Hz, 10.8 deg at 60 Hz, and 36 deg at 200 Hz. The benefit grows exactly where prediction and phase margin are hardest.

11. Exposure, ROI, and fast soundcard branch

Exposure trade

```
tau_exposure = T_exp/2
shot-noise standard deviation approximately proportional to 1/sqrt(T_exp)
```

Reducing exposure from 430 to 210 us recovers only about 110 us of midpoint delay, while halving photons and potentially increasing shot-noise standard deviation by sqrt(2). It helps only if the spot is bright enough and the measured end-to-end delay actually falls. The attenuation14 run should remain unchanged; exposure should be swept in a separate controlled test.

32x32 ROI under predictive control

The previous pure-PID 32x32 trial does not settle the predictive-control question. A historical delay reduction near 0.25 ms would move the run11 linear floor toward about 0.068 px Y / 0.104 px X if signal and noise were unchanged. But the smaller ROI brought a stray reflection within about 11 px, changes normalized CoM sensitivity, and requires a new pixel scale (15.5), Bode gain K, delay, and open-PSD bound.

Internal soundcard high-frequency branch

```
u_actuator = LPF[u_slow] + HPF[u_fast]
```

Audio mode	One 32-sample buffer	Implication
192 kHz	0.167 ms	promising if direct/exclusive and deterministic
96 kHz	0.333 ms	may still beat PiPlate mean ZOH
48 kHz	0.667 ms	unlikely to improve the current path

Because this would be an internal/onboard or PCIe soundcard rather than USB, the main risks are audio-stack buffering, codec group delay, asynchronous audio/camera clocks, and crossover phase. Use direct low-latency access, a fixed small buffer, complementary filters near 20 Hz, analog summing and protection, then measure the complete summed command-to-centroid Bode and latency. The fast branch is meaningful only if it reduces the measured actuator response delay, not merely the software write time.

Recommended hardware order

First finish run14. Next measure a low-exposure/ROI variant and the internal soundcard path independently. Favor the change that demonstrates at least 0.2-0.3 ms deterministic end-to-end reduction without raising the centroid noise floor. Only then combine the paths.

12. Data required for the true conditional MMSE

The present attenuation DAT files contain PSDs and summary statistics, not the original synchronized sequence. A PSD determines second-order linear prediction performance but cannot determine the full conditional distribution needed for a nonlinear conditional-MMSE value.

Telemetry	Why it is needed	Current availability
Raw 64x64 RAW8 frame	spot shape, flux, nonlinear centroid information	available before center_of_mass; not stored at full rate
Centroid Y/X	baseline predictor input and score	UDP 64732 full rate; PSD retained
Requested Y/X command	plant-aware prediction and leakage diagnosis	available inside FSP; not synchronized in DAT
Applied/clamped command	saturation and actuator-state reconstruction	pipeline state; needs recorder
Iteration/timestamp	align camera and actuator histories	AYLP header / recorder addition
Trip/ramp/phase flags	exclude transitions and faults	needs recorder fields

Measurement program

Experiment	Purpose
Dark frames, same exposure/gain	read/dark noise distribution
Stationary illuminated spot	centroid measurement floor and flux dependence
10-30 min open loop	disturbance distribution and nonstationarity
Small safe PRBS or multisine	joint plant, latency, and hysteresis identification
Closed-loop synchronized telemetry	out-of-sample command-aware prediction and safety analysis
Chronological train/validation/test	prevent leakage and quantify real conditional prediction error

Storage estimates

Centroids plus commands are only about 0.1 MB/s. Full 64x64 RAW8 frames are about 9.5 MB/s at 2310 Hz, or about 4.7 GB for 500 s. A practical recorder should always retain centroids/commands and capture raw frames in selected 30-60 s blocks.

Identifiability limit

The camera alone cannot perfectly separate true beam motion from camera measurement noise. Dark/static-spot calibration gives a useful empirical decomposition, but a rigorous physical beam-position bound would require an independent reference sensor. The existing hardware is nevertheless sufficient for a strong controller-visible conditional-MMSE estimate.

13. Recommended sequence after attenuation14

Priority	Action	Decision criterion
1	Interpret attenuation14 PDF/DAT and compute its own order128/order512 bounds	stable for full run; no clipping/trip; compare closed RMS and 120-400 Hz waterbed
2	Add synchronized centroid + requested/applied command recorder	time-aligned sequence with phase/ramp flags and no loop-rate loss
3	Run exposure/ROI latency-noise sweep	choose minimum predicted residual, not maximum frame rate
4	Bench-test internal soundcard branch electrically and optically	verified deterministic end-to-end saving >=0.2-0.3 ms
5	Implement joint X/Y command-aware subspace predictor in shadow mode	out-of-sample improvement over scalar FIR without active-model instability
6	Promote model only after constrained replay and stability checks	command/rate limits, validation gate, rollback controller

Preferred next control architecture

A joint X/Y command-aware data-driven subspace predictor feeding constrained MPC is the strongest practical successor. It uses [e_y,e_x,u_y,u_x] history, identifies a closed-loop subspace model without relying on fragile pseudo-open-loop reconstruction, computes updates in a shadow process, validates them, and swaps only stable controllers. PO4AO or another nonlinear policy may eventually exploit nonlinear/nonstationary structure, but its inference latency must be included in the same causal budget.

Performance expectations

Case	Y RMS	X RMS	Basis
Current run11 measured	0.127	0.218	completed hardware run
Current scalar linear floor	0.0769	0.1203	run11 PSD, 4.18 frames
Full path -0.5 ms floor	0.0584	0.0863	same PSD, 3.025-frame horizon
Exceptional practical hardware after latency improvement	0.065-0.080	0.095-0.115	engineering range, not a theorem

The most defensible statement is not that a particular model guarantees sub-0.1 px on both axes. It is that the measured disturbance contains enough predictable variance for sub-0.1 Y, while X needs either closer approach to the linear bound, lower latency, or useful extra information beyond the scalar centroid history.

14. References and workspace traceability

1. C. Kulcsar, H.-F. Raynaud, C. Petit, J.-M. Conan, and P. Viaris de Lesegno, 'Optimal control, observers and integrators in adaptive optics,' Optics Express 14, 7464-7476 (2006). https://doi.org/10.1364/OE.14.007464
2. C. Petit, J.-M. Conan, C. Kulcsar, H.-F. Raynaud, and T. Fusco, 'First laboratory validation of vibration filtering with LQG control law for adaptive optics,' Optics Express 16, 87-97 (2008). https://doi.org/10.1364/OE.16.000087
3. S. Meimon, C. Petit, T. Fusco, and C. Kulcsar, 'Tip-tilt disturbance model identification for Kalman-based control scheme: application to XAO and ELT systems,' JOSA A 27, A122-A132 (2010). https://doi.org/10.1364/JOSAA.27.00A122
4. C. Correia et al., 'Advanced vibration suppression algorithms in adaptive optics systems,' JOSA A 29, 185-194 (2012). https://doi.org/10.1364/JOSAA.29.000185
5. S. Y. Haffert et al., 'Data-driven subspace predictive control of adaptive optics for high-contrast imaging,' JATIS 7, 029001 (2021). https://doi.org/10.1117/1.JATIS.7.2.029001
6. J. Nousiainen et al., 'Laboratory demonstration of model-based reinforcement learning for adaptive optics,' JATIS (2024). https://arxiv.org/abs/2401.00242
7. J. Nousiainen et al., 'On-sky demonstration of reinforcement learning for adaptive optics control' (2026). https://arxiv.org/abs/2606.10771
8. J. Fowler et al., 'Ground control to major time-lag: on-sky results of data-driven predictive wavefront control at Keck Observatory' (2026). https://arxiv.org/abs/2606.20838

Primary workspace artifacts

Artifact	Role
devices/fsp.c, devices/fsp.h	controller implementation and state
doc/devices/fsp.md	controller model, parameters, and tuning rationale
contrib/attenuation_fsp.json	attenuation14 A/B configuration and output path
contrib/steering_fsp.json	current continuous steering configuration
devices/attenuation_test.c/.h	open/settle/closed acquisition and Welch report
test/fsp_wiener_bound.c	PSD-derived scalar linear prediction bound
Bode_and_testruns/attenuation10_lead_fc64.dat/.pdf	classical-loop baseline
Bode_and_testruns/attenuation11_fsp.dat/.pdf	completed full-band FSP result

Reproducibility note

All numeric PSD bounds in this report were recomputed from the DAT files using the same Wiener-Hopf method as test/fsp_wiener_bound.c. The report uses the current code/configuration snapshot and does not overwrite or reinterpret the still-running attenuation14 result.